Root-mean-square speed = 433 m s^{-1}

20 B

$$\frac{kq}{2.0} + \frac{k(2q)}{2.0} = \frac{3kq}{2.0} = 6740$$

The electric field at the mid-point due to the positive and negative charge are in the same direction.

$$\frac{kq}{2.0^2} + \frac{k(2q)}{2.0^2} = \frac{3kq}{4.0}$$

$$= \frac{6740}{2}$$

$$= 3370 \text{ N C}^{-1}$$